1) a)

When $x = 4$ then $t = \sqrt{4} = 2$

$$\therefore \pi \int_0^2 y^2 \frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t = \pi \int_0^2 (t^3)^2(2t)\,\mathrm{d}t = \pi \int_0^2 2t^7\,\mathrm{d}t = \pi\left[\frac{1}{4}t^8\right]_0^2 = \pi[(64) - 0]$$

$= 64\pi$ units^3

1) b)

When $x = 16$ then $t = 8^{\frac{1}{3}} = 2$

$$\therefore \pi \int_0^2 y^2 \frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t = \pi \int_0^2 (t^2)^2(6t^2)\,\mathrm{d}t = \pi \int_0^2 6t^6\,\mathrm{d}t = \pi\left[\frac{6}{7}t^7\right]_0^2 = \pi\left[\frac{768}{7} - 0\right]$$

$= \frac{768}{7}\pi$ units^3

1) c)

When $x = 28$ then $t = 27^{\frac{1}{3}} = 3$

$$\therefore \pi \int_0^3 y^2 \frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t = \pi \int_0^3 (t^2)^2(3t^2)\,\mathrm{d}t = \pi \int_0^3 3t^6\,\mathrm{d}t = \pi\left[\frac{3}{7}t^7\right]_0^3 = \pi\left[\frac{6561}{7} - 0\right]$$

$= \frac{6561}{7}\pi$ units^3

1) d)

$$\pi \int_0^{\frac{\pi}{2}} y^2 \frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t = \pi \int_0^{\frac{\pi}{2}} \left(\sqrt{\cos t}\right)^2(\cos t)\,\mathrm{d}t = \pi \int_0^{\frac{\pi}{2}} \cos^2 t\,\mathrm{d}t$$

Since $\cos 2A = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2} + \frac{1}{2}\cos 2A$ and so

$$\pi \int_0^{\frac{\pi}{2}} \cos^2 t\,\mathrm{d}t = \pi \int_0^{\frac{\pi}{2}} \left(\frac{1}{2} + \frac{1}{2}\cos 2t\right)\mathrm{d}t$$

$$= \pi\left[\frac{1}{2}t + \frac{1}{4}\sin 2t\right]_0^{\frac{\pi}{2}} = \pi\left[\left(\frac{\pi}{4} + 0\right) - (0 + 0)\right]$$

$= \frac{1}{4}\pi^2$ units^3

2) a)

$$\pi\int_0^1 x^2\frac{\mathrm{d}y}{\mathrm{d}t}\,\mathrm{d}t = \pi\int_0^1 \left(\sqrt{1-t^2}\right)^2(2t)\,\mathrm{d}t = \pi\int_0^1 (2t-2t^3)\,\mathrm{d}t$$

$$= \pi\left[t^2-\tfrac{1}{2}t^4\right]_0^1 = \pi\left[\left(1-\tfrac{1}{2}\right)-0\right] = \tfrac{1}{2}\pi \text{ units}^3$$

2) b)

When $y=1$ then $t = \arctan 1 = \frac{\pi}{4}$

$$\pi\int_0^{\frac{\pi}{4}} x^2\frac{\mathrm{d}y}{\mathrm{d}t}\,\mathrm{d}t = \pi\int_0^{\frac{\pi}{4}} (2\cos t)^2(\sec^2 t)\,\mathrm{d}t = \pi\int_0^{\frac{\pi}{4}} 4\,\mathrm{d}t$$

$$= \pi[4t]_0^{\frac{\pi}{4}} = \pi[\pi-0] = \pi^2 \text{ units}^3$$

2) c)

$$\pi\int_0^{\frac{\pi}{2}} x^2\frac{\mathrm{d}y}{\mathrm{d}t}\,\mathrm{d}t = \pi\int_0^{\frac{\pi}{2}} \left(\sin^2 t\sqrt{\cos t}\right)^2(\sin t)\,\mathrm{d}t = \pi\int_0^{\frac{\pi}{2}} \cos t\sin^5 t\,\mathrm{d}t$$

$$= \pi\left[\tfrac{1}{6}\sin^6 t\right]_0^{\frac{\pi}{2}} = \pi\left[\tfrac{1}{6}-0\right] = \tfrac{1}{6}\pi \text{ units}^3$$

3)

When $x = \mathrm{e}^2$ then $t = 4$

When $x = \mathrm{e}^3$ then $t = 6$

$$\therefore \pi \int_4^6 y^2 \frac{\mathrm{d}x}{\mathrm{d}t}\mathrm{d}t = \pi \int_4^6 \left(\sqrt{t-2}\right)^2 \left(\frac{1}{2}\mathrm{e}^{\frac{1}{2}t}\right)\mathrm{d}t = \frac{1}{2}\pi \int_4^6 \mathrm{e}^{\frac{1}{2}t}(t-2)\,\mathrm{d}t$$

Integration by Parts:

$$u = t - 2 \qquad \frac{\mathrm{d}v}{\mathrm{d}t} = \mathrm{e}^{\frac{1}{2}t}$$

$$\frac{\mathrm{d}u}{\mathrm{d}t} = 1 \qquad v = 2\mathrm{e}^{\frac{1}{2}t}$$

$$\frac{1}{2}\pi \int_4^6 \mathrm{e}^{\frac{1}{2}t}(t-2)\,\mathrm{d}t$$

$$= \frac{1}{2}\pi \left(\left[2\mathrm{e}^{\frac{1}{2}t}(t-2)\right]_4^6 - \int_4^6 2\mathrm{e}^{\frac{1}{2}t}\,\mathrm{d}t\right)$$

$$= \frac{1}{2}\pi \left[2\mathrm{e}^{\frac{1}{2}t}(t-2) - 4\mathrm{e}^{\frac{1}{2}t}\right]_4^6$$

$$= \frac{1}{2}\pi[(8\mathrm{e}^3 - 4\mathrm{e}^3) - (4\mathrm{e}^2 - 4\mathrm{e}^2)]$$

$= 2\pi\mathrm{e}^3$ units^3 as req.

4)

When $y = 1$ then $t = 1$

When $y = 9$ then $t = 3$

$$\therefore \pi \int_1^3 x^2 \frac{\mathrm{d}y}{\mathrm{d}t}\mathrm{d}t = \pi \int_1^3 \left(\sqrt{\ln|t+1|}\right)^2 (2t)\,\mathrm{d}t = \pi \int_1^3 2t \ln|t+1|\,\mathrm{d}t$$

Integration by Parts:

$$u = \ln|t+1| \qquad \frac{\mathrm{d}v}{\mathrm{d}t} = 2t$$

$$\frac{\mathrm{d}u}{\mathrm{d}t} = \frac{1}{t+1} \qquad v = t^2$$

$$\pi \int_1^3 t(\ln|t+1|)^2\,\mathrm{d}t$$

$$= \pi\left([t^2 \ln|t+1|]_1^3 - \int_1^3 \frac{t^2}{t+1}\mathrm{d}t\right)$$

But since, $\frac{t^2}{t+1} = \frac{t(t+1)-t}{t+1} = t - \frac{t}{t+1} = t - \frac{t+1-1}{t+1} = t - 1 + \frac{1}{t+1}$

$$\pi\left([t^2 \ln|t+1|]_1^3 - \int_1^3 \frac{t^2}{t+1}\mathrm{d}t\right) = \pi\left([t^2 \ln|t+1|]_1^3 - \int_1^3 \left(t - 1 + \frac{1}{t+1}\right)\mathrm{d}t\right)$$

$$= \pi\left[t^2 \ln|t+1| - \frac{1}{2}t^2 + t - \ln|t+1|\right]_1^3$$

$$= \pi\left[\left(9\ln 4 - \frac{9}{2} + 3 - \ln 4\right) - \left(\ln 2 - \frac{1}{2} + 1 - \ln 2\right)\right]$$

$$= \pi\left[\left(8\ln 4 - \frac{3}{2}\right) - \left(\frac{1}{2}\right)\right]$$

$$= \pi[16\ln 2 - 2]$$

$$= 2\pi(8\ln 2 - 1) \text{ units}^3 \text{ as req.}$$

5) a)

When $y = 0$ then $t = 1$, when $y = \ln 4$ then $t = 4$

$$\pi \int_1^4 x^2 \frac{\mathrm{d}y}{\mathrm{d}t}\mathrm{d}t = \pi \int_1^4 \left(\frac{4}{t}\right)^2 \left(\frac{1}{t}\right)\mathrm{d}t = \pi \int_1^4 16t^{-3}\,\mathrm{d}t$$

$$= \pi\left[-\frac{8}{t^2}\right]_1^4 = \pi\left[-\frac{8}{16} - (-8)\right] = \frac{15}{2}\pi \text{ cm}^3$$

5) b)

The model assumes the funnel has no thickness and that it perfectly follows the curve.

6) a)

$y = \sqrt{\sin t} = 0 \Rightarrow t = 0, \pi, \ldots$ so $k = \pi$

6) b)

$\therefore \pi \int_0^{\pi} y^2 \frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t = \pi \int_0^{\pi} \left(\sqrt{\sin t}\right)^2 (\sin t)\,\mathrm{d}t = \pi \int_0^{\pi} \sin^2 t\,\mathrm{d}t$

Since $\cos 2A = 1 - 2\sin^2 A$ then $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos 2A$ and so

$$\pi \int_0^{\pi} \sin^2 t\,\mathrm{d}t = \pi \int_0^{\pi} \left(\frac{1}{2} - \frac{1}{2}\cos 2t\right)\mathrm{d}t$$

$$= \pi \left[\tfrac{1}{2}t - \tfrac{1}{4}\sin 2t\right]_0^{\pi}$$

$$= \pi \left[\left(\tfrac{1}{2}\pi - 0\right) - (0 - 0)\right]$$

$= \frac{1}{2}\pi^2$ units^3

7)

$$\pi \int_0^{k} x^2 \frac{\mathrm{d}y}{\mathrm{d}t}\,\mathrm{d}t = \pi \int_0^{k} \left(\sqrt{t}\right)^2 \mathrm{e}^t\,\mathrm{d}t = \pi \int_0^{k} t\mathrm{e}^t\,\mathrm{d}t$$

Consider $\int t\mathrm{e}^t\,\mathrm{d}t$ only:

Integration by Parts:

$u = t$	$\frac{\mathrm{d}v}{\mathrm{d}t} = \mathrm{e}^t$
$\frac{\mathrm{d}u}{\mathrm{d}t} = 1$	$v = \mathrm{e}^t$

$$\therefore \int t\mathrm{e}^t\,\mathrm{d}t = t\mathrm{e}^t - \int \mathrm{e}^t\,\mathrm{d}t = t\mathrm{e}^t - \mathrm{e}^t\ (+c)$$

So

$$\pi \int_0^{k} t\mathrm{e}^t\,\mathrm{d}t\ = \pi[t\mathrm{e}^t - \mathrm{e}^t]_0^k$$

$$= \pi[(k\mathrm{e}^k - \mathrm{e}^k) - (0 - 1)]$$

$$= \pi\left((k-1)\mathrm{e}^k + 1\right)$$

Equate: $\pi(3\mathrm{e}^4 + 1) = \pi\left((k-1)\mathrm{e}^k + 1\right)$

$3\mathrm{e}^4 = (k-1)\mathrm{e}^k$

$\therefore k = 4$

8) a)

$u = \sin\theta$ so $\frac{\mathrm{d}u}{\mathrm{d}\theta} = \cos\theta$

Since $\cos^3\theta = \cos\theta\,(1 - \sin^2\theta)$ then

$\int \cos\theta\,(1 - \sin^2\theta)\,\mathrm{d}\theta = \int (1 - u^2)\,\mathrm{d}u = u - \frac{1}{3}u^3 + c = \sin\theta - \frac{1}{3}\sin^3\theta + c$

8) b)

$\pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} y^2 \frac{\mathrm{d}x}{\mathrm{d}\theta}\,\mathrm{d}\theta = \pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\cos\theta)^2(\cos\theta)\,\mathrm{d}\theta = \pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^3\theta\,\mathrm{d}\theta = \pi\left[\sin\theta - \frac{1}{3}\sin^3\theta\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$

$= \pi\left[\left(1 - \frac{1}{3}\right) - \left(-1 + \frac{1}{3}\right)\right] = \frac{4}{3}\pi$ units^3

9) a)

$xy = (\cosh t - \sinh t)(\cosh t + \sinh t)$

$= \cosh^2 t - \sinh^2 t$

$= 1$

So that $y = \frac{1}{x}$

Since $\cosh t \geq 1$ for any t and for $t \leq 0$ then $\sinh t \leq 0$

So $x = \cosh t - \sinh t \geq 1 - 0 = 1$

$\therefore$ for $t \leq 0$ and $x \geq 1$

9) b)

$\lim_{a\to\infty} \pi \int_1^a y^2\,\mathrm{d}x = \lim_{a\to\infty} \pi \int_1^a x^{-2}\,\mathrm{d}x = \lim_{a\to\infty} \pi\left[-\frac{1}{x}\right]_1^a = \lim_{a\to\infty} \pi\left[-\frac{1}{a} - (-1)\right]$

$= \pi(0 + 1) = \pi$

So $V = \pi$ units^3

10) a)

When $t = 0$, $x = a$ and $y = 0 \Rightarrow a = 2$

When $t = 2$, $x = a\mathrm{e}^{-2}$ and $y = 4b \Rightarrow b = \frac{3}{4}$

$a = 2$ and $b = \frac{3}{4}$

10) b)

$$\pi\int_0^2 x^2\frac{\mathrm{d}y}{\mathrm{d}t}\mathrm{d}t = \pi\int_0^2 (2\mathrm{e}^{-t})^2\left(\frac{3}{2}t\right)\mathrm{d}t = 6\pi\int_0^2 t\mathrm{e}^{-2t}\,\mathrm{d}t$$

Integration by Parts:

$$u = t \qquad \frac{\mathrm{d}v}{\mathrm{d}t} = \mathrm{e}^{-2t}$$

$$\frac{\mathrm{d}u}{\mathrm{d}t} = 1 \qquad v = -\frac{1}{2}\mathrm{e}^{-2t}$$

$$\therefore \int t\mathrm{e}^{-2t}\,\mathrm{d}t = -\frac{1}{2}t\mathrm{e}^{-2t} + \frac{1}{2}\int \mathrm{e}^{-2t}\,\mathrm{d}t = -\frac{1}{2}t\mathrm{e}^{-2t} - \frac{1}{4}\,\mathrm{e}^{-2t}(+c)$$

And so

$$6\pi\int_0^2 t\mathrm{e}^{-2t}\,\mathrm{d}t = 6\pi\left[-\tfrac{1}{2}t\mathrm{e}^{-2t} - \tfrac{1}{4}\,\mathrm{e}^{-2t}\right]_0^2 = 6\pi\left[\left(-\mathrm{e}^{-4} - \tfrac{1}{4}\,\mathrm{e}^{-4}\right) - \left(0 - \tfrac{1}{4}\right)\right] = 4.28\text{ cm}^3$$

10) c)

The model assumes the funnel has no thickness and that it perfectly follows the curve.

11) a)

Maximum of $x = \sqrt{t^2 - 1}$ is when $t = 3$

$\therefore$ Max diameter is $2 \times \sqrt{3^2 - 1} = 4\sqrt{2} = 5.7$ cm (to the nearest mm)

11) b)

Maximum of $y = t + \ln t$ is when $t = 3$

But minimum of $y = t + \ln t$ is when $t = 1$ so $y = 1$

$\therefore$ Max height is $3 + \ln 3 - 1 = 3.1$ cm (to the nearest mm)

11) c)

$$\pi\int_1^3 x^2\frac{\mathrm{d}y}{\mathrm{d}t}\mathrm{d}t = \pi\int_1^3 \left(\sqrt{t^2-1}\right)^2\left(1 + \frac{1}{t}\right)\mathrm{d}t = \pi\int_1^3 t^2 + t - 1 - \frac{1}{t}\mathrm{d}t$$

$$= \pi\left[\tfrac{1}{3}t^3 + \tfrac{1}{2}t^2 - t - \ln t\right]_1^3 = \pi\left[\left(\tfrac{1}{3}\times 3^3 + \tfrac{1}{2}\times 3^2 - 3 - \ln 3\right) - \left(\tfrac{1}{3} + \tfrac{1}{2} - 1 - 0\right)\right]$$

$$= 30.1\text{ cm}^3$$

11) d)

Mass = density × volume = $2.5 \times 30.0 = 75$ g

12) a)

$\frac{dy}{dt} = \frac{2te^t - t^2e^t}{e^{2t}} = te^{-t}(2-t)$

Max when $\frac{dy}{dt} = 0$

So $te^{-t}(2-t) = 0 \Rightarrow t = 0$ or $t = 2 \therefore t = 2$

$x = a = \frac{2}{2} = 1$ and $y = \frac{2^2}{e^2}$

$P\left(1, \frac{4}{e^2}\right)$

12) b)

When $x = 1$, $t = 2$, when $x = 3$, $t = \frac{2}{3}$

$$\pi \int_{2}^{\frac{2}{3}} y^2 \frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t = -\pi \int_{\frac{2}{3}}^{2} \left(\frac{t^2}{\mathrm{e}^t}\right)^2 (-2t^{-2})\,\mathrm{d}t = 2\pi \int_{\frac{2}{3}}^{2} t^2 \mathrm{e}^{-2t}\,\mathrm{d}\theta$$

Integration by Parts:

$$u = t^2 \qquad \frac{\mathrm{d}v}{\mathrm{d}t} = \mathrm{e}^{-2t}$$

$$\frac{\mathrm{d}u}{\mathrm{d}t} = 2t \qquad v = -\frac{1}{2}\mathrm{e}^{-2t}$$

$$\therefore \int t^2 \mathrm{e}^{-2t}\,\mathrm{d}t = -\frac{1}{2}t^2 \mathrm{e}^{-2t} + \int t\,\mathrm{e}^{-2t}\mathrm{d}t$$

Integration by Parts:

$$u = t \qquad \frac{\mathrm{d}v}{\mathrm{d}t} = \mathrm{e}^{-2t}$$

$$\frac{\mathrm{d}u}{\mathrm{d}t} = 1 \qquad v = -\frac{1}{2}\mathrm{e}^{-2t}$$

$$\therefore \int t^2 \mathrm{e}^{-2t}\,\mathrm{d}t = -\frac{1}{2}t^2 \mathrm{e}^{-2t} + \left(-\frac{1}{2}t\mathrm{e}^{-2t} + \frac{1}{2}\int \mathrm{e}^{-2t}\,\mathrm{d}t\right)$$

$$= -\frac{1}{2}t^2 \mathrm{e}^{-2t} - \frac{1}{2}t\mathrm{e}^{-2t} - \frac{1}{4}\mathrm{e}^{-2t}(+c)$$

$$\therefore 2\pi \int_{\frac{2}{3}}^{2} t^2 \mathrm{e}^{-2t}\,\mathrm{d}t = \pi\left[-\frac{1}{2}t^2 \mathrm{e}^{-2t} - \frac{1}{2}t\mathrm{e}^{-2t} - \frac{1}{4}\mathrm{e}^{-2t}\right]_{\frac{2}{3}}^{2}$$

$$= 2\pi\left[\left(-2\mathrm{e}^{-4} - \mathrm{e}^{-4} - \tfrac{1}{4}\mathrm{e}^{-4}\right) - \left(-\tfrac{2}{9}\mathrm{e}^{-\frac{4}{3}} - \tfrac{1}{3}\mathrm{e}^{-\frac{4}{3}} - \tfrac{1}{4}\mathrm{e}^{-\frac{4}{3}}\right)\right] = 2\pi\left[\left(-\tfrac{13}{4}\mathrm{e}^{-4}\right) - \left(-\tfrac{29}{36}\mathrm{e}^{-\frac{4}{3}}\right)\right]$$

$$= \pi\left(-\frac{13}{2\mathrm{e}^4} + \frac{29}{18\mathrm{e}^{\frac{4}{3}}}\right)$$

$$= \pi\left(\frac{29}{18\mathrm{e}^{\frac{4}{3}}} - \frac{13}{2\mathrm{e}^4}\right) \text{ units}^3 \text{ as req.}$$

13)

The point P is when $x = \ln 13$

So $\ln(t-3) = \ln 13 \Rightarrow t = 16$

When $x = 0$, $\ln(t-3) = 0 \Rightarrow t - 3 = \mathrm{e}^0 \therefore t = 4$

So

$$\pi \int_4^{16} y^2 \frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t = \pi \int_4^{16} \left(\frac{8}{\sqrt{t}}\right)^2 \left(\frac{1}{t-3}\right)\mathrm{d}t = \pi \int_4^{16} \frac{64}{t(t-3)}\,\mathrm{d}t$$

Partial fractions:

$$\frac{64}{t(t-3)} \equiv \frac{A}{t} + \frac{B}{t-3}$$

$$64 = A(t-3) + Bt$$

Let $t = 0$: $\quad 64 = -3A$

Let $t = 3$: $\quad 64 = 3B$

$$\therefore \frac{64}{t(t-3)} = \frac{64}{3(t-3)} - \frac{64}{3t}$$

$$\therefore \pi \int_4^{16} \frac{64}{t(t-3)}\,\mathrm{d}t = \pi \int_4^{16} \left(\frac{64}{3(t-3)} - \frac{64}{3t}\right)\mathrm{d}t =$$

$$= \pi \left[\frac{64}{3}\ln|t-3| - \frac{64}{3}\ln|t|\right]_4^{16} = \pi \left[\frac{64}{3}\ln\left|\frac{t-3}{t}\right|\right]_4^{16} = \frac{64}{3}\pi\left(\ln\left(\frac{13}{16}\right) - \ln\left(\frac{1}{4}\right)\right)$$

$$= \frac{64}{3}\pi \ln\left(\frac{13}{4}\right)$$

Also:

When $x = \ln 13$, $y = \frac{8}{\sqrt{16}} = 2$

$$y - 2 = \frac{2}{\ln 13}(x - \ln 13) \Rightarrow y = \frac{2}{\ln 13}x$$

$$\pi \int_0^{\ln 13} \left(\frac{2}{\ln 13}x\right)^2 \mathrm{d}x = \frac{4}{(\ln 13)^2}\pi \int_0^{\ln 13} x^2\,\mathrm{d}x = \frac{4}{(\ln 13)^2}\pi \left[\frac{1}{3}x^3\right]_0^{\ln 13}$$

$$= \frac{4}{(\ln 13)^2}\pi\left(\frac{1}{3}(\ln 13)^3 - 0\right) = \frac{4\ln 13}{3}\pi$$

$$\frac{64}{3}\pi \ln\left(\frac{13}{4}\right) - \frac{4\ln 13}{3}\pi = \frac{4}{3}\pi\left(16\ln\left(\frac{13}{4}\right) - \ln 13\right)$$

So the total volume is:

$$\frac{4}{3}\pi\left(16\ln\left(\frac{13}{4}\right) - \ln 13\right) \text{ units}^3$$